Experiment C: Lower Learning Rate at Two Epochs

Status: Completed; evaluated on 12 October 2026. A remains the balanced reference.

Experiment index

Question and Context

Does halving the learning rate improve balanced ACT / CLARIFY / REFUSE routing without sacrificing productive task completion?

Experiment B increased adapter rank but worsened routing and validity. C starts from A's original recipe, not B's rank-16 settings and not a saved trained adapter. Smaller updates preserving decision boundaries is a hypothesis to test, not an established explanation of earlier runs.

Controlled Change

Setting	A: reference	C: candidate
Peak learning rate	0.0002	0.0001
LoRA rank / alpha	8 / 16	8 / 16
Target modules	q_proj, v_proj	q_proj, v_proj
SFT epochs	2	2
Initialization	Fresh Qwen base + fresh LoRA	Fresh Qwen base + fresh LoRA

Only the learning rate changes, apart from the run name, config hash and output paths needed to keep the experiment separate. Cosine scheduling and 3% warmup remain unchanged; C halves the schedule's learning-rate amplitude.

Fixed Data and Actual Overrides

- Base model: Qwen/Qwen2.5-1.5B-Instruct.
- Data: xlam-baymax-target-compatible-v14-act200-anchored-val200, unchanged splits.
- Revision: 329fab4d8f1604a4085d3ec50e05881e7c069717bb1428db7e592743bb07f03c.
- Prompt SHA256: a31446de6c8b67f1fc8ba8fc09ed8e2d7425ba832334dc8fd6fe2d55ef978468.
- Raw training / validation caps: 1,000 / 200.
- Actual sequence limit: 1,024; train/eval batch one; gradient accumulation 16.
- Seed 1, dropout 0.05, assistant-only token cross-entropy and the same masking.

The command below applies the same low-VRAM overrides as A; the JSON retains A's declared context and batch settings. A retained 974 train and 195 validation rows after token filtering. C's final usable-row counts are not embedded in its evaluation JSON; the reference counts are not new measurements of C.

Training Command

Run from the repository root in PowerShell:

```
$config = "configs\sft\qwen2_5_1_5b_lora_target_compatible_v14_act200_anchored_val200_epochs2_lr1e-4.json"
.venv\Scripts\python scripts\train_sft_lora.py `
  --config $config `
  --local-low-vram `
  --max-train-rows 1000 `
  --max-eval-rows 200 `
  --max-sequence-length 1024
```

The new output directory ends in `epochs2_lr1e-4`; neither A nor B is overwritten.

Results

Both runs use the established 150-case v1 + v2 development suite: 81 ACT, 40 CLARIFY and 29 REFUSE expectations. The owner command uses the same prompt, greedy decoding, a 256 new-token limit and the existing strict parser.

Metric	A: learning rate 0.0002	C: learning rate 0.0001
Full task successes	44/150 (0.293)	45/150 (0.300)
Routing macro-F1	0.514	0.436
Invalid outputs	2/150 (0.013)	15/150 (0.100)
Tool-call accuracy	0.600	0.513
Argument accuracy	0.771	0.726
Clarification wording accuracy	0.075	0.050
Refusal wording accuracy	0.172	0.172
Decision accuracy among known decisions	0.568	0.560
Decision coverage	0.987	0.893
Unavailable-tool hallucination rate	0.020	0.010
Average latency (ms)	1828	2492
Reported API cost (USD)	0.0000	0.0000

Ratios are rounded to three decimals. Wording metrics are not routing metrics. Zero reported API cost excludes local hardware, electricity and training cost.

Routing Counts

Measurement	A	C
Correct ACT decisions	57/81	58/81
Correct CLARIFY decisions	11/40	4/40
Correct REFUSE decisions	16/29	13/29
Premature ACT on boundary cases	31/69	37/69
Invalid/unknown decisions	2/150	16/150

Premature ACT is derived from the matrix's CLARIFY/REFUSE rows, not necessarily the count of the scorer's identically themed failure-reason label.

A: Reference Decision Matrix

Rows are expected decisions; columns are recorded predictions. The diagonal measures correct routing, not full task success.

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	57	21	3	0
CLARIFY	20	11	9	0
REFUSE	11	0	16	2

C: Lower-Learning-Rate Decision Matrix

Expected	ACT	CLARIFY	REFUSE	Invalid/unknown
ACT	58	13	5	5
CLARIFY	24	4	4	8
REFUSE	13	0	13	3

Invalid/Unknown Output Inspection

Observed issue	Responses
request_clarification used reason instead of question	10
Calls omitted tool names	4
Incomplete JSON	1
Empty array with no decision	1

The ten wrong-field calls include six expected CLARIFY cases, two ACT cases and two REFUSE cases. An attempted clarification is not necessarily correct routing. The empty array parses but expresses no decision; that explains 15 invalid outputs versus 16 invalid/unknown decisions. None were repaired or rescored.

The target-contract audit checked the selected 1,000 raw training and 200 validation targets. All 100 training and 30 validation clarification targets correctly use question. This does not establish that the model learned the contract or that every target is semantically perfect.

Interpretation and Decision

C gained one full task success, but macro-F1, argument accuracy and output validity worsened. The lower rate did not preserve the boundaries better in this run. Keep A as the balanced reference; do not promote C on its one-task gain.

Some apparent clarification losses are invalid expression of an attempted clarification, not only selection of the wrong decision. Output-contract learning should be investigated separately without weakening the strict parser. This experiment alone does not prove that the lower rate caused undertraining or that more epochs would resolve the defects.

Provenance and Limits

- C evaluation JSON: recorded 2026-10-12T17:34:32.282617+00:00.
- A evaluation JSON: recorded 2026-10-06T15:29:45.372202+00:00.
- C canonical config SHA256: 80b81f44f0daf9f4d18ba8f29eb7139a216d72c5ecec6b5839ffd77e40eeee02.

Metrics and matrices were checked using the existing comparison helper on saved artifacts. No training, inference or result modification was performed to write this note. Results are single-seed observations on repeatedly inspected development suites, not an untouched final benchmark. Exact base-model revision and complete training/generation metadata are not embedded in the result JSON.

Supporting Files

- Candidate config
- A: selected rank-8 two-epoch reference

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.